

RESEARCH ARTICLE

Adaptive Learning for Congestion Control in 6LoWPAN Networks with Dynamic Priority

Jiasheng Pan | Jiayang Zhou | Li Jin

Global College, Shanghai Jiao Tong University, Shanghai, China |

Correspondence: Li Jin (li.jin.sjtu@edu.cn)

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ABSTRACT

The IPv6 over low-power wireless personal area network protocol, tailored for low-power and lossy networks, is critical for integrating Wireless Sensor Networks into the Internet of Things ecosystem. However, the protocol's inherent constraints often lead to network congestion. In particular, when nodes possess dynamic priorities, it complicates congestion control mechanisms, an issue that current research has yet to fully explore. To tackle this, we propose Adaptive Learning for Congestion Control (ALCC), a novel method combining game theory and reinforcement learning to optimize congestion management. Our method employs a noncooperative game model to adjust packet sending rates to maximize aggregate transmission in traffic control phase. Concurrently, a cooperative game framework manages resources, factoring in dynamic node priorities and environmental conditions. This is enhanced by the Multi-Agent Proximal Policy Optimization algorithm for adaptive resource allocation. Simulation results demonstrate that our method outperforms existing congestion control methods. Compared to existing congestion control methods, our method achieves lower packet loss rates, higher throughput, and a substantially improved fairness, ensuring both efficiency and equitable resource distribution.

1 | Introduction

The Internet of Things (IoT) has emerged as a transformative paradigm, seamlessly interconnecting devices such as sensors, actuators, and appliances to form a smart network ecosystem [1][2][3]. Wireless Sensor Networks (WSNs) are central to this framework, providing low-power, cost-effective devices that collect and transmit data to the Internet. The IPv6 over Low-Power Wireless Personal Area Network (6LoWPAN) protocol stack [4], standardized by the Internet Engineering Task Force, enables this integration for resource-constrained devices, operating over IEEE 802.15.4 with the Routing Protocol for Low-Power and Lossy Networks (RPL) [5]. However, incorporating 6LoWPAN into IoT networks introduces challenges, particularly in congestion management—an area frequently explored in research [6][7]—that compromise the Quality of Service vital for IoT applications.

To mitigate congestion in 6LoWPAN networks, two primary strategies are utilized: resource control and traffic control. Resource control identifies alternative, noncongested paths for packet transmission, whereas traffic control adjusts data sending rates to alleviate network-wide congestion [8]. An effective solution must navigate a complex trade-off between maximizing overall throughput and maintaining a fair Weighted Fairness Index (WFI), especially when sensor nodes possess dynamic priorities based on application urgency. However, existing game-theoretic models often fail to adapt to these dynamic priorities, leading to either the "starvation" of low-priority nodes or suboptimal resource utilization in the presence of high-priority bursts. In response, this paper considers the following research question: Can we devise a method that ensures high network throughput while maintaining a high WFI across nodes with dynamic priorities?

Existing studies address network congestion primarily through traffic control [8][9] and resource control [10][11]. From the

traffic control perspective, early works like ESRT [12] regulate source reporting frequencies to maintain event-to-sink reliability rather than end-to-end reliability, while Woo et al. [13] utilize an adaptive rate control mechanism to balance originating and route-through traffic. Furthermore, the Fusion framework [14] integrates hop-by-hop backpressure, source rate limiting, and prioritized medium access to throttle traffic during overload. Conversely, resource control methods alleviate congestion by dynamically increasing network capacity or altering routing topologies. For instance, HTAP [15] creates hierarchical alternative paths to bypass congested nodes, TARA [16] activates dormant nodes based on topology analysis to absorb traffic bursts, and Siphon [17] deploys multi-radio virtual sinks to redirect overload traffic onto a secondary network. While these heuristic methods effectively alleviate localized buffer overflows in the short term, their structurally isolated design—treating micro-level rate allocation and macro-level topology adjustment as disjoint tasks—often leads to suboptimal network-wide resource utilization and severe route flapping, highlighting the critical need for a tight cross-layer coupling mechanism to ensure stability under dynamic priorities.

To achieve distributed fairness and mathematically optimal rate allocation without centralized overhead, game-theoretic models have been extensively applied to WSN and IoT congestion control. Al-Kashoash et al. [18][19] pioneered this direction by developing a noncooperative game model for traffic control, which was later augmented with Multi-Attribute Decision Making (MADM) in the OHCA algorithm. Similarly, Chowdhury et al. [20] proposed NGECC, integrating energy constraints within a noncooperative game to optimize resource utilization. As highlighted by comprehensive surveys in this domain [21][22], game theory has proliferated into various network layers to formulate strategic interactions among resource-constrained nodes. For instance, recent advancements like GTCC [23] leverage potential game theory specifically for RPL-enabled networks, allowing child nodes to autonomously select optimal parents to scatter traffic flows and mitigate the ping-pong effect. Despite establishing solid mathematical foundations for proportional fairness and distributed decision-making, these game-theoretic models exhibit a critical limitation: they inherently rely on static priority assumptions and rigid equilibrium conditions. In highly dynamic 6LoWPAN environments where localized hazards trigger sudden bursts of critical data, node priorities shift rapidly. Current noncooperative games struggle to converge under such dynamic priority conditions, leading to unfair bandwidth allocation and the potential starvation of time-critical alerts. This reveals a pressing need to structurally embed dynamic priority awareness into the micro-level game formulation, ensuring that equilibrium outcomes instantly adapt to the shifting urgency of sensor data.

In recent years, Machine Learning, particularly Reinforcement Learning (RL), has emerged as a powerful tool for cross-layer network optimization by enabling nodes to autonomously learn adaptive routing and congestion control policies. For instance, Naeem et al. [24] proposed a model-free adaptive actor-critic deep reinforcement learning framework based on Software-Defined Networking (SDN) and fuzzy neural networks to address multipath congestion control in heterogeneous IoT environments. Focusing on QoS provisioning, Muthanna et al. [25] utilized a Soft Actor-Critic (SAC) reinforcement learning model to dynamically enforce transmission policies and optimize multi-hop

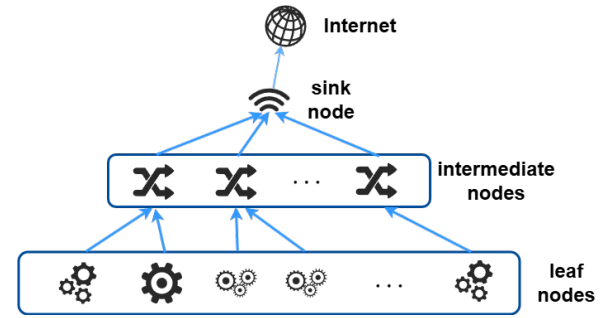


FIGURE 1 | Schematic diagram of network topology.

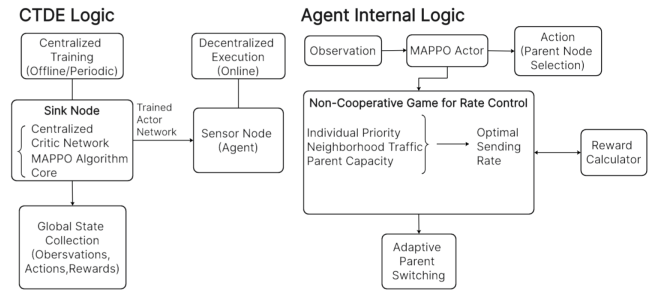


FIGURE 2 | Structural diagram of optimization framework.

routing in LoRa networks. Furthermore, to tackle the real-time constraints of highly dynamic edge networks, He et al. [26] introduced RTHop, a decentralized multi-agent deep reinforcement learning (MADRL) approach that leverages real-time Markov decision processes to achieve adaptive hop-by-hop routing and alleviate network congestion. Hou et al. [27], using reinforcement learning for both control aspects, achieved promising results in fixed-priority networks, but their model lacks generalizability to dynamic settings. Although these advanced RL and MADRL models significantly enhance the adaptivity of network protocols, they generally lack generalizability to environments with rapidly shifting node priorities and fail to structurally couple macro-level routing decisions with micro-level noncooperative rate constraints. This highlights a critical need for a deeply integrated hybrid framework that natively embeds micro-level game-theoretic payoffs into macro-level RL rewards to simultaneously guarantee dynamic weighted fairness and topological stability.

To bridge this gap, we propose a hierarchical optimization framework that integrates microscopic traffic control with macroscopic resource management. The structural diagram is shown in Figure 2, which illustrates the Centralized Training for Decentralized Execution logic and agent internal logic.

In the traffic control phase, we formulate a novel non-cooperative game as a localized optimization problem. By incorporating average priority and homogeneous penalty terms—directly tied to the parent node's aggregate capacity—into the payoff design, our approach achieves a Nash equilibrium that sustains high throughput for heterogeneous nodes sharing a common parent. Specifically, the inclusion of the priority metric ensures that bandwidth allocation strictly adheres to the varying urgency of data streams, preventing time-critical packets from being starved

by routine traffic. Meanwhile, the penalty term acts as a rigorous constraint to proactively avert localized buffer overflows.

In the resource control phase, we leverage the Markovian properties of the problem and apply the Multi-Agent Proximal Policy Optimization (MAPPO) algorithm [28] for adaptive parent node selection. This stage treats the equilibrium payoffs derived from the traffic control phase as the primary input for the reward function, thereby creating a tight coupling between rate allocation and routing decisions. Through this cross-layer synergy, the macro-level routing behavior is fundamentally driven by the micro-level rate satisfaction of each individual node. If a parent node becomes persistently congested—resulting in degraded payoffs in the traffic game—the MAPPO agents are intrinsically incentivized to discover and shift traffic to underutilized paths, ensuring that dynamic topology adjustments strictly serve the ultimate goal of maximizing network-wide, priority-aware throughput.

To address the rapid priority shifts inherent in WSNs, we embed individual priority, neighborhood average priority, and real-time congestion metrics into the MAPPO agents' observation space, enabling proactive, context-aware routing. We construct the reinforcement learning reward to maximize the sum of adjusted payoffs from the traffic control phase, ensuring that macro-level topology changes directly serve micro-level rate allocation. Crucially, to protect energy-constrained WSN nodes from the heavy signaling overhead of route flapping, we incorporate a switching cost constraint. This design effectively suppresses unnecessary topology oscillations, achieving both a superior WFI and long-term network stability.

We validated this approach through extensive network simulations, testing various configurations with differing priority arrangements. Compared to baselines of OHCA and NGECC, our framework exhibits superior throughput, packet delivery ratios, and a significant enhancement in WFI.

In summary, the contributions of this paper are as follows.

1. A Novel Non-cooperative Game Framework: A traffic control model that boosts throughput while ensuring weighted fairness for nodes with dynamic priorities.
2. A Stability-Aware MAPPO Resource Control: An RL-based parent selection mechanism that optimizes global social welfare while penalizing excessive topology oscillations.
3. Simulation-based validation demonstrating marked improvements in throughput, delivery ratios, and WFI compared to existing methods.

The rest of this paper is organized as follows. Section 2 describes the network setup and presents the modeling and solution of the traffic control problem using a noncooperative game. Section 3 addresses the resource control problem through cooperative game modeling and reinforcement learning. Section 4 provides details on the implementation of the algorithm and presents simulation experiments comparing ALCC performance with that of other methods. Finally, Section 5 offers concluding remarks.

2 | Network Setup and Traffic Control Modeling

As introduced in Section 1, the effective management of congestion requires a solid foundation of network topology. This section first establishes the 6LoWPAN network environment using the RPL protocol. Subsequently, we formulate the microscopic traffic control problem as a non-cooperative game to derive the optimal sending rates for leaf nodes.

2.1 | Network Setup

We use adopted RPL routing protocol to construct the network topology of 6LoWPAN network. The network consists of three types of nodes: a sink node, intermediate nodes, and leaf nodes. The sink node, as the root, acts as the gateway to external networks. Intermediate nodes forward data between the sink and leaf nodes, which are nonforwarding end devices. The overall network structure is illustrated in Figure 1, where leaf nodes may possess varying priorities based on the applications they host.

The DODAG construction process begins with the sink node broadcasting a DODAG Information Object (DIO) control message. This message contains critical information, such as the ID and rank value of the node. Neighboring nodes that receive the DIO message calculate their own rank based on the objective function, decide whether to join the DODAG, and, if so, send a Destination Advertisement Object (DAO) message back to the sink to confirm their membership. After joining, these nodes broadcast their own DIO messages to their neighbors, propagating the topology formation throughout the network. This process continues until the DIO messages reach the leaf nodes. When a node receives multiple DIO messages from different neighbors, it selects the neighbor with the best rank as its parent. If a node fails to receive a DIO message within a specified time, it can proactively send a DODAG Information Solicitation (DIS) message to request DIO messages from its neighbors.

In this model, we consider a parent node, denoted as I_k , associated with n competing leaf nodes, $L_1^{I_k}, L_2^{I_k}, \dots, L_n^{I_k}$. Each leaf node behaves selfishly, seeking to maximize its own sending rate while contending with its neighbors for the limited resources of the shared parent node. The complete set of leaf nodes across m parent nodes is represented as $L_i^{I_k}$, where $i = 1, 2, \dots, n$ indexes the leaf nodes under a given parent node, and $k = 1, 2, \dots, m$ indexes the parent nodes.

2.2 | Microscopic Traffic Control Problem Formulation

In the traffic control phase, we address the optimization of sending rates for leaf nodes in a 6LoWPAN network, where these nodes compete for the limited resources of a shared parent node. To model this noncooperative dynamic, we employ a noncooperative game framework, which is particularly suited to capturing the selfish behavior of the nodes and addressing the challenges

posed by incomplete information sharing in such networks. The problem formulation rests on the following assumptions:

1. **Maximum Sending Rate:** Each leaf node L_i has a maximum sending rate, denoted by $\xi_i^{\max}$.
2. **Priority Levels:** Each leaf node hosts applications of varying importance, which assigns a priority level p_i to node L_i based on the types of applications it supports.

The noncooperative game is formally defined as $G = (N, \Phi_i, F_i)$ for $i = 1, 2, \dots, n$, with the following components:

Players: We have n players $L_1, L_2, \dots, L_n$, each of them share the same parent node I_k .

Strategies: The strategy space for each leaf node L_i is $\Phi_i = [0, \xi_i^{\max}]$, reflecting the range of possible sending rates. The joint strategy space for all players is $\Phi = \prod_{i=1}^n \Phi_i$.

Payoffs: The payoff function for each player L_i is $F_i : \Phi \rightarrow \mathbb{R}$, quantifying the balance between maximizing the sending rate and minimizing associated costs. The payoff function F_i comprises three key components:

1. **Utility Function:** For a parent node I_k with a maximum sending rate ξ_{out} , each leaf node L_i aims to maximize its sending rate. To ensure concavity and the existence of a Nash equilibrium [29], the utility function is defined as:

$$U_i = \log \left(\frac{\omega_5 \xi_i}{\xi_{out}} + 1 \right), \quad (1)$$

where ω_5 is a weighting parameter. The logarithmic utility function inherently captures the diminishing marginal returns of bandwidth allocation. In the context of 6LoWPAN, this logarithmic structure is physically motivated by the principle of proportional fairness. It ensures that while high-priority nodes are granted larger shares of the parent node's capacity (ξ_{out}), low-priority nodes are protected from complete starvation, thereby maintaining a robust and equitable sensing coverage.

2. **Cost Functions:** Two cost functions are considered:
Congestion Cost Function: the congestion cost function (C_i) shows the congestion situation of the parent node I_k of the leaf node L_i . Since it is influenced by all leaf nodes of I_k , we denote

$$C_i = n \frac{\sum_{i=1}^n \beta_i \xi_i}{\xi_{out}}, \quad (2)$$

where β_i is 1 - the probability of channel loss. By incorporating the channel loss parameter β_i , the cost accurately reflects the effective traffic arriving at the parent, making the model robust against the inherent unreliability of 6LoWPAN links.
Priority Cost Function: the priority cost function (P_i) denotes the relative importance of the applications hosted by leaf node L_i , which can be expressed as

$$P_i = \left(p_i - \bar{p} + \frac{\omega_4}{\omega_3} \right) \frac{\xi_i}{\xi_{out}}, \quad (3)$$

where the p_i is the priority of the player L_i and $\bar{p}$ is the average priority of all leaf node of I_k . We introduce additional parameters, ω_3 and ω_4 , to achieve the desired differentiation

in sending rates for nodes with varying priorities.

The design of the priority cost function embeds a mechanism of localized competitive awareness. Instead of relying solely on an absolute priority metric, the incorporation of the differential term $(p_i - \bar{p})$ evaluates a node's urgency relative to its immediate siblings sharing the same parent. This is crucial for network stability during burst events. When the regional average priority $\bar{p}$ surges, the cost penalty adaptively suppresses excessively aggressive transmissions, preventing channel collapse.

Combining these components, the payoff function for each leaf node L_i is expressed as

$$F_i = \omega_1 \log \left(\frac{\omega_5 \xi_i}{\xi_{out}} + 1 \right) - \omega_2 n \frac{\sum_{i=1}^n \beta_i \xi_i}{\xi_{out}} - \omega_3 (p_i - \bar{p}) \frac{\xi_i}{\xi_{out}} - \omega_4 \frac{\xi_i}{\xi_{out}}, \quad (4)$$

where $\omega_1, \omega_2, \omega_3, \omega_4$ are weighting coefficients that balance the utility and cost terms.

2.3 | Solution

The optimization problem for each leaf node L_i is formulated as a constrained nonlinear program:

$$\begin{aligned} \text{Max} \quad & F_i(\xi_i, \xi_{-i}) \\ \text{s.t.} \quad & \xi_i \geq 0, \xi_i \leq \xi_i^{\max}, \end{aligned} \quad (5)$$

where $i = 1, 2, \dots, n$, and ξ_{-i} represents the strategies of all other players. To solve this optimization problem, we employ the Lagrangian method by introducing Lagrange multipliers χ_i and v_i for the constraints $\xi_i \geq 0$ and $\xi_i \leq \xi_i^{\max}$ respectively. The Lagrangian is defined as

$$\mathcal{L}_i(\xi_i, \chi_i, v_i) = F_i(\xi_i, \xi_{-i}) + \chi_i \xi_i + v_i (\xi_i^{\max} - \xi_i). \quad (6)$$

The Karush-Kuhn-Tucker (KKT) conditions for optimality are

$$\begin{aligned} \chi_i, v_i &\geq 0, \xi_i \geq 0, \xi_i^{\max} - \xi_i \geq 0, \\ \frac{\partial F_i}{\partial \xi_i} + \chi_i - v_i &= 0, \chi_i \xi_i = 0, \\ v_i (\xi_i^{\max} - \xi_i) &= 0. \end{aligned}$$

Given that the payoff function is concave and the inequality constraints admit an interior feasible point, the KKT conditions can be applied to determine the optimal sending rate ξ_i^* . The solution is

$$\xi_i^* = \begin{cases} 0, & \text{if Condition 1 holds,} \\ \xi_i^{\max}, & \text{if Condition 2 holds,} \\ \frac{w_1 \xi_{out}}{w_2 n \beta_i + w_3 (p - \bar{p}) + w_4} - \frac{\xi_{out}}{w_5}, & \text{otherwise} \end{cases} \quad (7)$$

where two conditions respectively are

$$\text{Condition 1 : } w_2 n \beta_i + w_3 (p - \bar{p}) + w_4 \geq w_1 w_5,$$

$$\text{Condition 2 : } w_2 n \beta_i + w_3 (p - \bar{p}) + w_4 \leq \frac{w_1 \xi_{out}}{\xi_i^{max} + \frac{\xi_{out}}{w_5}}.$$

For traffic control, the complexity is $\mathcal{O}(1)$ per node, as it leverages a closed-form solution derived from a non-cooperative game, requiring only basic arithmetic operations. For resource control using MAPPO, while the centralized training at the edge is more intensive, the decentralized execution on sensor nodes involves only a lightweight neural network forward pass with a constant complexity of $\mathcal{O}(H^L)$ (where H is the number of hidden units and L is the number of layers). This asymmetric design ensures that the real-time operational burden remains minimal for resource-constrained 6LoWPAN nodes.

3 | Resource Control Modeling

In the resource control phase, we address the dynamic parent node selection problem for leaf nodes in a 6LoWPAN network, extending the game-theoretic traffic control framework into a multi-agent reinforcement learning setting. RPL employs a default objective function, such as OF0 [5] or MRHOF [30], to construct the DODAG. Building on this foundation, certain methods determine the objective function combining specific metrics, such as the expected number of transmissions (ETX), energy, and buffer occupancy (BO) to guide parent node selection. In contrast to these prior approaches, we model this phase as a Partially Observable Markov Decision Process (POMDP). Within this framework, decentralized leaf nodes select parent nodes under conditions of partial observability, leveraging traffic control outcomes to optimize overall network performance. The integration of game theory and MAPPO ensures a cohesive approach to both phases. Here we employ a cooperative game framework, since the traffic control strategy ensures individual node payoffs while a strategic node distribution significantly enhances overall network performance.

3.1 | Macroscopic Resource Control Problem Formulation

Players: We have N players $L_1, L_2, \dots, L_N$, and each of them need to select a parent node from $I = (I_1, \dots, I_M)$ according to their own state.

State space: The global state $s(t) \in \mathcal{S}$ at time t includes

- $\xi_{out}(t)$: sending rates of intermediate nodes,
- $BO(t)$: buffer occupancies,
- $\bar{p}(t)$: average priorities per intermediate node,
- $p(t) = (p^1(t), \dots, p^N(t))$: priorities of the leaf nodes,

- $ETX(t) = (ETX^1(t), \dots, ETX^N(t))$: the expected number of transmissions from leaf nodes to all intermediate nodes,
- $I(t) = (I^1(t), \dots, I^N(t))$: current parent assignments.

Each agent L_i observes a local state $o_i(t) = (\xi_{out}(t), BO(t), \bar{p}(t), p^i(t), ETX^i(t), I^i(t))$, capturing only a subset of the global state, thus embodying partial observability.

Action Space: Each agent L_i selects a parent node $a_i(t) \in I$, with joint actions forming $\mathcal{A} = I^N$.

Payoffs: The payoff (or reward) for each leaf node L_i at time t , denoted $R_i(t)$, comprises traffic control payoff and switching cost. For traffic control payoff, according to the definition of the payoff function F_i in the traffic control phase, we define the reward function in each single traffic control part as R_1 , where

$$R_i^1(t) = \frac{\omega_1 \xi_{out}}{\omega_5} \log \left(\frac{\omega_5 \xi_i}{\xi_{out}} + 1 \right) - \omega_2 n - \omega_3 (p_i - \bar{p}) \frac{\xi_i}{\xi_{out}} - \omega_4 \frac{\xi_i}{\xi_{out}}. \quad (8)$$

The payoff function is built upon Equation 4. This formular creates a tight cross-layer coupling. By treating the microscopic Nash equilibrium payoff as the primary RL reward, MAPPO intrinsically drives the agents to maximize the global social welfare of the network. To encourage agents to prioritize higher throughput, we introduce a weighting factor ξ_{out}/ω_5 to the utility term. Additionally, we simplify the congestion cost function by replacing the original term $n(\sum_{i=1}^n \beta_i \xi_i)/\xi_{out}$ with a constant $\omega_2 n$. This adjustment reduces computational complexity, facilitating learning in the model while maintaining the essence of congestion-related costs.

For switching cost, frequent parent reselection not only consumes extra energy but also introduces severe topology oscillations (route flapping). To suppress this instability, we define a switching penalty cost c :

$$R_i^2(t) = \begin{cases} 0 & \text{if } \delta(t) = I_i(t), \\ -c & \text{otherwise.} \end{cases} \quad (9)$$

This penalty acts as a crucial regularization term, providing the routing topology with 'inertia' to prevent nodes from blindly switching due to marginal short-term utility gains.

Thus, the total reward for L_i at time t is

$$R_i(t) = R_i^1(t) + R_i^2(t). \quad (10)$$

Objective: The agents aim to maximize the expected cumulative discounted shared reward over time, which is

$$E \left[\sum_{i=1}^N \sum_{t=0}^{\infty} \gamma^t R_i(t) \right],$$

where γ is the discount factor.

Algorithm 1 Simplified MAPPO for Resource Control in 6LoWPAN

```

Initialize shared actor network  $\pi_\theta$  and
centralized critic network  $V_\phi$ 
Set hyperparameters: discount factor  $\gamma$ , GAE
parameter  $\lambda$ , clipping parameter  $\epsilon$ , entropy
coefficient  $\beta$ , number of epochs  $K$ , batch size  $B$ ,
trajectory length  $T$ 
for each iteration do
    Set  $\theta_{\text{old}} \leftarrow \theta$ 
    Initialize trajectory set  $\tau$ 
    for  $t = 0$  to  $T - 1$  do
        Collect observations  $o(t)$  from state  $s(t)$ 
        for each agent  $i = 1$  to  $N$  do
            Sample action  $a_i(t) \sim \pi_{\theta_{\text{old}}}(\cdot | o_i(t))$ 
            Compute probability  $p_i(t) = \pi_{\theta_{\text{old}}}(a_i(t) | o_i(t))$ 
        end for
        Form joint action  $a(t) = (a_1(t), \dots, a_N(t))$ 
        Execute  $a(t)$ :
            Update parent assignments  $I(t+1) = a(t)$ 
            Solve traffic control game for optimal
            rates  $\xi_i^*$  by Equation 7 and compute rewards  $R_i(t)$  by
            Equation 10
            Store  $(s(t), o(t), a(t), p(t), R(t), s(t+1))$  in  $\tau$ 
        end for
        Compute shared rewards  $R_{\text{sum}}(t) = \sum_{i=1}^N R_i(t)$  for
        each  $t$ 
        Compute advantages  $A(t)$  using GAE
        Compute returns  $\hat{R}(t) = A(t) + V_\phi(s(t))$ 
        for each epoch  $k = 1$  to  $K$  do
            Sample mini-batch of size  $B$  from  $\tau$ 
            for each sample  $b$  do
                for each agent  $i$  do
                    Compute ratio  $r_{\theta,i}^b = \frac{\pi_\theta(a_i^b | o_i^b)}{p_i^b}$ 
                    Compute clipped term  $c_i^b = \text{clip}(r_{\theta,i}^b, 1 - \epsilon, 1 + \epsilon) A^b$ 
                    Compute  $L_i^b = \min(r_{\theta,i}^b A^b, c_i^b) + \beta S[\pi_\theta](o_i^b)$ 
                end for
            end for
            Update  $\theta$  to maximize  $\frac{1}{B} \sum_{b=1}^B \left[ \frac{1}{N} \sum_{i=1}^N L_i^b \right]$ 
            Update  $\phi$  to minimize  $\frac{1}{B} \sum_{b=1}^B (V_\phi(s^b) - \hat{R}^b)^2$ 
        end for
    end for
end for

```

3.2 | MAPPO Solution

To solve this cooperative game, we employ MAPPO, a multi-agent reinforcement learning algorithm tailored for cooperative settings. MAPPO leverages a centralized training with decentralized execution approach, making it ideal for this problem: during training, a centralized critic uses global information to guide learning, while during execution, agents act based solely on their local observations.

As an Actor Critic algorithm, MAPPO's architecture for this problem consists of two primary components: Shared Actor Network and Centralized Critic.

Shared Actor Network: Each agent L_i uses a policy network $\pi_\theta(a_i | o_i)$, parameterized by θ , to map its local observation $o_i(t)$ to a probability distribution over actions (parent nodes). This enables decentralized execution, as agents do not need real-time access to others' observations or actions.

Centralized Critic : A value network $V_\phi(s)$, parameterized by ϕ , estimates the expected future shared reward from the global state $s(t)$. The critic's access to $s(t)$ during training ensures that policies are optimized for the collective good. The actor updates its policy by maximizing a clipped surrogate objective:

$$L(\theta) = \mathbb{E} [\min(r_\theta A, \text{clip}(r_\theta, 1 - \epsilon, 1 + \epsilon) A)] + \beta \mathbb{E} [S[\pi_\theta]] \quad (11)$$

where

$$r_\theta = \frac{\pi_\theta(\mathbf{a} | \mathbf{o})}{\pi_{\theta_{\text{old}}}(\mathbf{a} | \mathbf{o})}. \quad (12)$$

Here $A_i^{(k)}$ is computed using the GAE method, S is the policy entropy, and β is the entropy coefficient hyperparameter.

And the critic is trained to minimize the mean squared error between its predictions and the actual returns

$$L(\phi) = \mathbb{E} [(V_\phi(s) - \hat{R})^2] \quad (13)$$

where $\hat{R}_i$ is the discounted reward-to-go.

Both the actor and critic parameters are optimized using the Adam optimizer [31], which adapts learning rates for efficient convergence. And the detailed description of the algorithm can be viewed at Algorithm 1.

3.3 | Complexity Analysis

After training the MARL model using MAPPO, the inference phase involves decentralized execution where each leaf node (agent) independently selects a parent node based on its local observations. Each agent's policy network (actor) maps a local observation $o_i(t)$ to a probability distribution over M parent nodes. For a neural network with L hidden layers, input dimension d_{in} , hidden layer dimensions d_h , and output dimension $d_{\text{out}} = M$. The computational complexity of a single forward pass per agent is dominated by matrix multiplications, expressed as: $\mathcal{O}(d_{\text{in}} \cdot d_h + (L - 1) \cdot d_h^2 + d_h \cdot d_{\text{out}})$. Given that $d_{\text{out}} = M$ and d_{in} typically scales linearly with M (e.g., $d_{\text{in}} \propto M$), the complexity can be approximated as $\mathcal{O}(M \cdot d_h + (L - 1) \cdot d_h^2)$.

4 | Performance Evaluation

This section evaluates the ALCC algorithm's performance under dynamic priority scenarios through extensive experiments. The result demonstrates that ALCC outperforms baseline protocols in terms of increasing throughput, and enhancing the Weighted Fairness Index (WFI). Additionally, ablation studies confirm that the integrated switching cost constraint is vital for suppressing

topology oscillations and maintaining long-term network stability.

4.1 | Algorithm Implementation

In the ALCC algorithm, the leaf node will flexibly combine the two control methods to improve its payoff. Initially, each leaf node transmits data at a maximum rate of ξ_{max} to its current parent node ρ_k in DODAG. When the total incoming data rate ξ_{total} exceeds the outgoing rate ξ_{out} at any intermediate node, the parent nodes in the DODAG initiate congestion control by sending congestion_DIO messages to their child nodes. Upon receiving these messages, the leaf nodes employ the MAPPO algorithm to decide whether to select a new parent node. If a leaf node decide to change parent, it sends a join_DAO message to the new parent node and a leave_DAO message to the previous parent node. After receiving DAO messages from the leaf nodes, the parent node updates its average priority and the total number of connected leaf nodes, then broadcasts a topology_DIO message containing this updated information. Upon receiving the topology_DIO message, the leaf nodes adjust their sending rates according to Equation 7.

To effectively manage this process, it is essential to estimate the sending rate ξ_{out} for each parent node. Due to some variables include the number of active nodes, ξ_{out} experiences minor fluctuations over time. To address this, we utilize Brown's simple exponential smoothing model [32] to estimate the actual maximum service rate of each intermediate node as follows:

$$\xi_{out}(t+1) = \eta \cdot \xi_{out}(t) + (1-\eta) \cdot \xi_{out}(t-1) \quad (14)$$

where η is the smoothing factor.

4.2 | Simulation Setup

We conducted simulations using Python 3.10 on WSL2, with each node implemented as a container via Docker. By configuring matching IP addresses and ports, we enabled communication between nodes. The simulated 6LoWPAN network comprises one sink node, three intermediate nodes, and ten leaf nodes.

To evaluate the effectiveness of the ALCC method, we also simulated the previously mentioned OHCA and NGECC approaches and compared the performance of all three methods. OHCA is a hybrid congestion control algorithm that integrates Gray Relational Analysis for optimal path selection with the Network Utility Maximization framework to dynamically adjust data transmission rates. NGECC utilizes non-cooperative game theory to determine the optimal data transfer rates for source nodes by simultaneously accounting for buffer overflow, channel occupancy loss, and node energy consumption.

In the simulation, we configured the following parameters: for the resource control phase, the MAPPO algorithm was employed, with both the Actor and Critic networks comprising two fully connected layers followed by a softmax layer. For the traffic control

phase, the weights were assigned as $\omega_1 = 15$, $\omega_2 = 9$, $\omega_3 = 12.8$, $\omega_4 = 11.52$, $\omega_5 = 12.8$ and the smoothing factor $\eta = 0.4$. For the resource control phase, switching cost $c = 1$. The maximum sending rate ξ_{max}^i for all nodes i was uniformly set to 12. The weighting coefficients ω_1 to ω_5 are determined based on literature review. Initially, the base values are adopted from the non-cooperative game model in [20] to ensure physical consistency with the standard 6LoWPAN channel and buffer characteristics. However, to accommodate the dynamic priority requirements and the weighted fairness objectives proposed in [19], these coefficients are further adjusted to reach maximum throughput. The buffer size of the intermediate nodes was established at 8. For additional parameters related to channel loss and buffer loss, we adopted the settings outlined in [20].

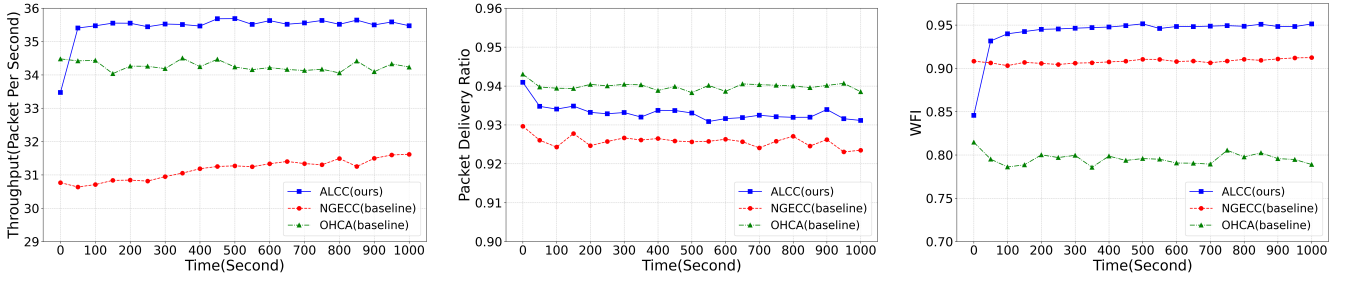
TABLE 1 | Leaf Node Connections for Different Methods

Parent Node	NGECC	OHCA	ALCC
0	Node 0: 1	Node 1: 1	Node 3: 1
	Node 1: 1	Node 3: 1	Node 5: 3
	Node 6: 3	Node 5: 3	Node 8: 3
	Node 9: 3	Node 6: 3	Node 9: 3
1	Node 2: 1	Node 2: 1	Node 0: 1
	Node 3: 1	Node 8: 3	Node 4: 3
	Node 7: 3		Node 6: 3
			Node 7: 3
2	Node 4: 3	Node 0: 1	Node 1: 1
	Node 5: 3	Node 4: 3	Node 2: 1
	Node 8: 3	Node 7: 3	
		Node 9: 3	

4.3 | Simulation Results

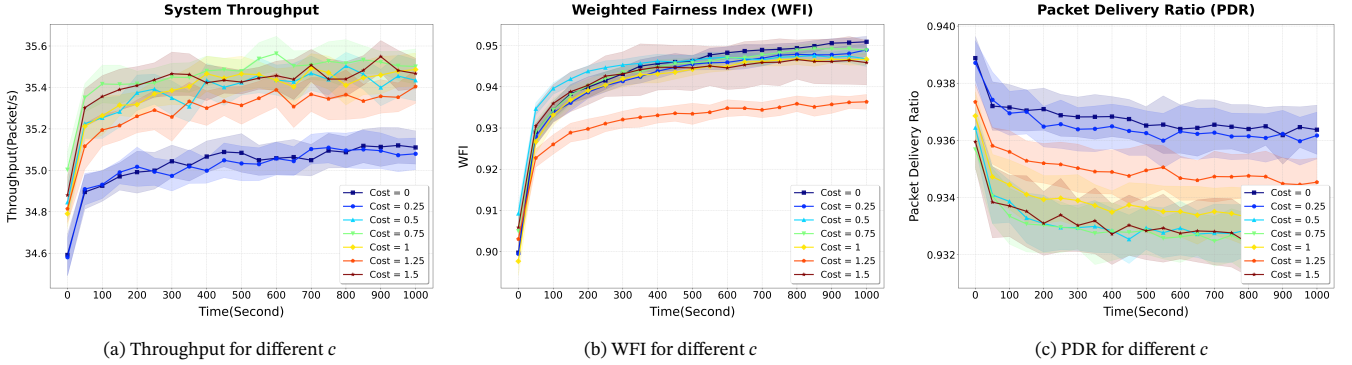
To provide an intuitive understanding of the impact of different methods on resource control, Table 1 illustrates the network topology after 500 seconds for various algorithms, displaying the leaf node indexes and their corresponding priorities under each parent node. Initially, intermediate node 1 was connected to leaf nodes 1, 2, 7, and 10; intermediate node 2 to leaf nodes 3, 4, and 8; and intermediate node 3 to leaf nodes 5, 6, and 9. As observed in the table, in an environment prone to congestion, ALCC demonstrates a node distribution that reflects a more rational allocation of priorities compared to OHCA and NGECC. Unlike ALCC, OHCA and NGECC do not explicitly incorporate priority distribution into their resource control mechanisms. Subsequent experiments will offer a more quantitative analysis of the performance differences among these methods.

To evaluate the overall performance of the congestion control algorithms, we conducted 500 independent trials for each method. In each trial, node priorities were assigned randomly to reflect diverse network conditions. Additionally, to simulate dynamic



(a) Comparison of throughput of different algorithms at the sink node. (b) Comparison of packet delivery ratio of different algorithms. (c) Comparison of weighted fairness index of different algorithms.

FIGURE 3 | Comparison of different aspects with baseline algorithms, illustrating the superior behavior of our algorithm.



(a) Throughput for different c (b) WFI for different c (c) PDR for different c

FIGURE 4 | Comparison between different c , illustrating that $c = 1$ is a reasonable choice

network behavior, we randomly selected one node every 100 seconds and altered its priority, mimicking the exit and entry of nodes..

Figure 3a illustrates the throughput at the sink node for the three algorithms. The results indicate that ALCC achieves higher throughput than both OHCA and NGECC. This advantage stems from the ability of leaf nodes under ALCC to select parent nodes at an optimal frequency, enabling prompt responses to changing network conditions. In contrast, OHCA and NGECC rely on simpler resource control logic, which may hinder leaf nodes from effectively reselecting intermediate nodes during complex congestion scenarios.

The packet delivery ratio serves as a critical metric for evaluating the success rate of packet transmission from leaf nodes. It is defined as the ratio of packets successfully received at the sink node to the total packets sent by all leaf nodes. As depicted in Figure 3b, ALCC exhibits a higher delivery ratio compared to previous methods, such as NGECC, which models traffic control as a noncooperative game. This improvement is attributable to the homogeneous term in Equation 4 of ALCC, which enhances coordination among nodes and mitigates packet loss. Nevertheless, due to the type of game, the delivery ratio of ALCC is 0.4% lower than that of OHCA.

Another key metric, the Weighted Fairness Index (WFI), measures the fairness of throughput allocation among leaf nodes based on their assigned priorities [19]. The WFI is calculated as

follows.

$$WFI = \frac{\left(\sum_{i=1}^n t_i \times p_i\right)^2}{n \sum_{i=1}^n (t_i \times p_i)^2} \quad (15)$$

where t_i is the throughput of the leaf node I_i . The WFI values presented in Figure 3c are averaged over 500 trials under varying priority configurations. The results reveal that ALCC achieves a higher WFI than both OHCA and NGECC. This superiority arises because ALCC explicitly considers node priorities when leaf nodes select parent nodes, accounting for the differing sending rates ξ_{out} of intermediate nodes. Consequently, ALCC ensures a more equitable distribution of resources across the network, aligning throughput allocation with node priorities more effectively than its counterparts.

Finally, we compute the average values of three critical metrics of three methods. To enable a standardized comparison, we normalize these values for each metric by assigning a score of 1.00 to the method exhibiting the highest performance. The performance of the remaining methods is then expressed as a ratio relative to this maximum value. The normalized results are presented in Table 2.

The superior WFI performance of ALCC shown in Table 2 is directly attributable to the intelligent parent reselection visualized in Table 1. Unlike OHCA, which relies on static metrics, ALCC's MAPPO agent proactively offloads high-priority traffic to underutilized intermediate nodes.

TABLE 2 | Normalized Analysis Results (Best Performance = 1.00)

Method	WFI	Throughput	Delivery Ratio
NGECC	0.9604	0.8792	0.9854
OHCA	0.9108	0.9759	1.0000
ALCC	1.0000	1.0000	0.9929

To explicitly validate the necessity of the switching penalty within our MAPPO reward design, we conducted a targeted ablation study. We conducted simulation with different switching costs c in Equation 9. These simulations were evaluated against the standard ALCC under identical conditions. Results are shown in Figure 4.

From Figure 4, we can conclude that reasonable switch cost can prevent aggressive switching, thereby increasing the overall throughput. The absence of a switching penalty leads to a phenomenon known as route flapping. They aggressively switch parent nodes to chase short-term utility gains in the rate allocation game. While excessive switch penalty can degrade WFI due to suppressing switching. In addition, PDR also decreased due to switch penalty. Switch cost $c = 1$ is a reasonable choice to balance these factors.

5 | Conclusion

In this paper, we considered the dynamic priority congestion control problem in 6LoWPAN networks. First, we address the traffic control problem by enabling leaf nodes to maximize their individual payoff functions. Building on the outcomes of the traffic control phase, we employ the MAPPO algorithm to ensure a high WFI while sustaining high throughput of leaf nodes. We refer to the proposed algorithm as ALCC. Extensive simulations demonstrate that ALCC substantially enhances both the WFI and throughput while achieving a packet delivery ratio comparable to, or marginally higher than, existing methods, such as those in [19] and [20].

For future work, we plan to incorporate additional factors, such as energy consumption, into the MAPPO framework during the resource control phase and adjust the payoff function accordingly. Furthermore, improving the robustness of the algorithm in scenarios where node properties—such as the number of nodes—are dynamic represents a valuable direction for further research.

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